Cabin Systems – General Data

1. General

The Cabin Passenger Address System, (if installed) is part of the audio intergration system, refer to 23-50-00, Audio Integration – General Data. The Cabin Passenger Address System allows the flight compartment crew to communicate with the passengers in the cabin.

The Cabin Passenger Address System consists of a dedicated NAT 245 speaker amplifier used to drive six speakers mounted in the cabin roof.

2. Description

A. Cabin Passenger Address System (NAT 245)

When the audio control panel PA is selected and keyed, the NAT 245 drives six 8 ohm cabin speakers. The NAT 245 speaker amplifier is mounted on the middle shelf in the forward avionics rack. With the pilot and co-pilot's audio controls on PA, the pilot's microphone has priority if both the pilot and co-pilot's PTT switches are pushed at the same time.

28 Volt DC power is protected by a 5 amp CABIN PA (5M) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The Cabin Passenger Address System interfaces to the following aircraft systems:

1) Audio Control Panel System (refer to 23-53-00, Audio Control Panel – General Data).

3. Adjustment/Test

A. Function Test Cabin Passenger Address System (NAT 245)

1) General

These procedures are for the function test of the Cabin Passenger Address System (NAT 245).

2) Function Test

a) Ensure that CABIN PA (5M) circuit breaker is closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical systems to initialize.

d) On the audio control panel, adjust SPKR/PA switch to select PA mode.

e) Key the PTT from the pilot's position. Verify audio is heard in the cabin.

f) Key the PTT from the co-pilot's position. Verify audio is heard in the cabin.

g) Key the PTT from the pilot and co-pilot's position. Verify that the audio from the pilot is heard in the cabin.

h) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Audio Control Panel PA Adjustment

1) General

These procedures are for the adjustment of the audio level of the cabin passenger address speakers.

2) Adjustment

a) Ensure that CABIN PA (5M) circuit breaker is closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical systems to initialize.

d) On the audio control panel, adjust SPKR/PA switch to select PA mode.

e) Key the PTT from the pilot's position. Verify audio is heard in the cabin.

f) Evaluate the audio level.

g) Remove the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 4.A.2).

h) Adjust the pilot's audio panel PA potentiometer adjustment on the top of the unit.

i) Install the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 4.A.2).

j) Test the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 3.A.2).

k) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTERGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the Cabin Passenger Address System (NAT 245)

1) General

These procedures are for the removal and installation of the Cabin Passenger Address System (NAT 245).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open CABIN PA (5M) circuit breaker on the avionics circuit breaker panel and fit with a safety clip.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Remove top and middle cover attached to bulkhead station 37.21.

e) Disconnect the electrical connector on the front of the unit.

f) Remove the (4) screws securing the unit to the forward avionics shelf.

g) Remove unit from aircraft.

Note Stow connectors as required.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the CABIN PA (5M) circuit breaker on the avionics circuit breaker panel is opened and fitted with a safety clip.

c) Align the units mounting holes to the forward avionics shelf.

d) Secure the unit to the avionics shelf with mounting (4) screws.

e) Reconnect connector to the unit. Ensure connections are secure.

f) Remove the safety clips and close CABIN PA (5M) circuit breaker on the avionics circuit breaker panel.

g) Test the Cabin Passenger Address System as per Para 3.A.

5. Troubleshooting

A. Cabin Passenger Address System (NAT 245)

1) If the Cabin Passenger Address System (NAT 245) is deemed inoperative check the following:

a) Ensure CABIN PA (5M) circuit breaker is closed.

b) Visually inspect the NAT 245 unit for damaged wires or contacts.

c) Visually inspect the cabin speakers for damaged wires or contacts.

d) Ensure that the audio NAT 245 and cabin speaker connectors are properly mated and secure.

e) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

f) If repair or service is required, send to appropriate rated avionics repair facility.

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